GLOBAL AI HACKATHON · PROBLEM STATEMENT 06

LumaSuri

Best-before dates for AI-written claims.

Tavily live verification **Bifrost LLM gateway** **Answers + whole repos**

luma (old) + *suri* (examine): examining what has gone old

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THE PROBLEM

AI answers freeze time, and then they spread.

“Node.js 18 is the current LTS.”
Now: Node.js 26 (Lithium)

“Use openai.Completion.create with text-davinci-003.”
Model retired, SDK rewritten

“Heroku has a free tier for hobby apps.”
Removed

“React 18 is the latest version.”
Now: React 19

Why it hurts

- LLMs answer from a training cutoff, with full confidence.
- Their answers get copied into **READMEs, code comments, docs and support bots**.
- Nobody re-checks a sentence after it ships. Facts expire silently.
- Fact-checkers say "true or false today". They don't say **when to check again**.

THE SOLUTION

Every claim gets a score, a source and an expiry date.

0–100

Freshness score

Combines support, evidence recency and source authority. Every verdict quotes the exact sentence from a live source.

Best-before date

Based on the source's **real release cadence** and how fast that kind of fact decays. It tells you when to re-check.

− / +

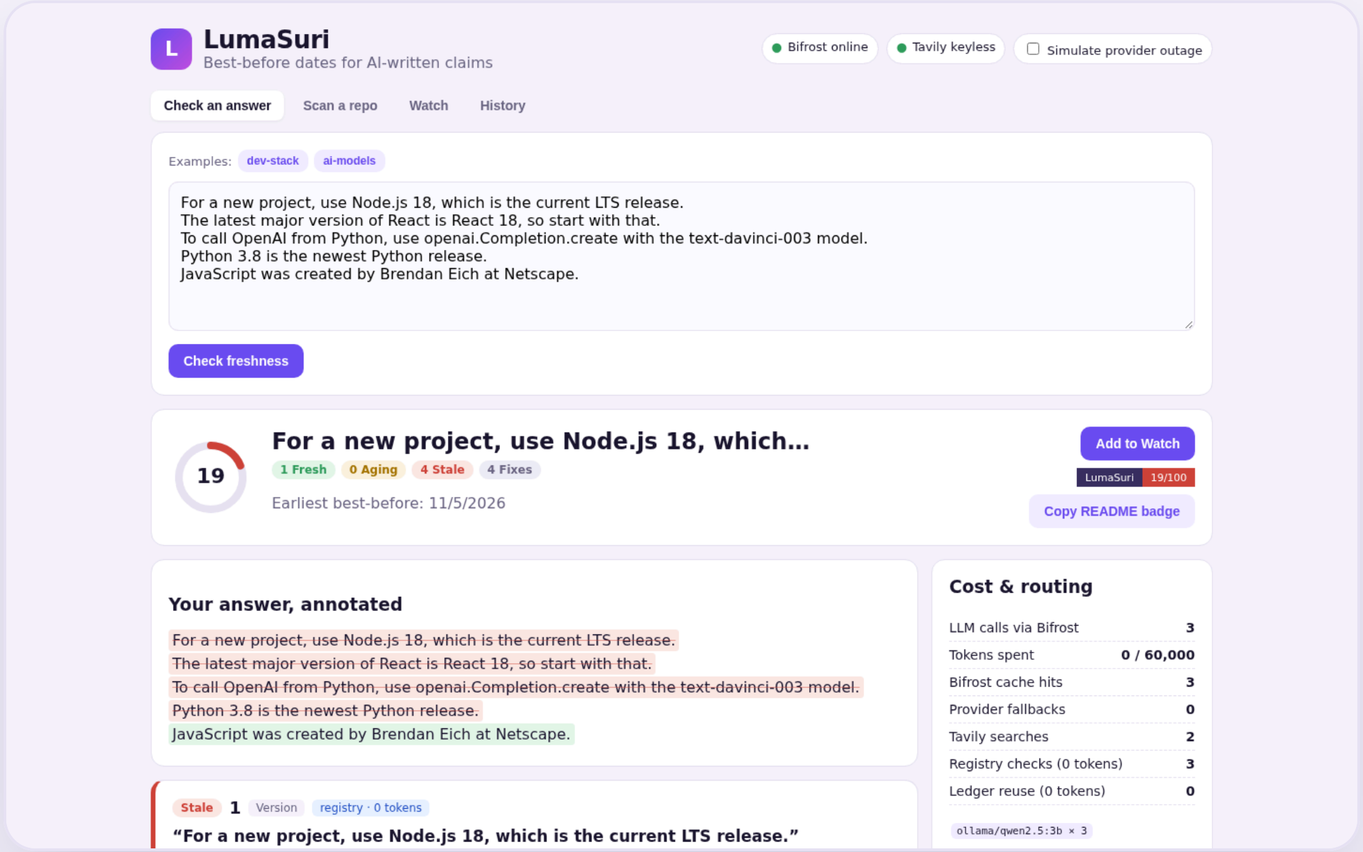
Fix diff

Stale claims come with a one-line fix for the exact file and line. You review it; LumaSuri never edits files itself.

LumaSuri Watch keeps saved answers and repos. When a claim passes its best-before date, Watch re-verifies it and shows **what changed since you saved it**.

WORKING PROTOTYPE

Paste an answer, or scan a repo.



Web app, CLI and an OpenAI-compatible guard proxy.

- Answer annotated in green / amber / red
- Claim cards: verdict, quoted evidence, official-source badge, fix diff
- Best-before date with the reason for it
- Cost & routing panel from Bifrost metadata
- `lumasuri scan . --fail-under 60` works as a CI check

TECHNICAL ARCHITECTURE

Cheapest check first. LLM only when needed.

1 · Candidate filter
Regex finds time-sensitive lines in answers, docs, comments and package.json
0 tokens

→

2 · Extract claims
Small model returns category, subject, value and line
Bifrost

→

3 · Ledger + registries
Reuse verdicts that haven't expired. npm · PyPI · nodejs.org · endoflife.date
0 tokens

→

4 · Live search
Official domains first, extracted chunks
Tavily

→

5 · Evidence-locked verdict
Judge must quote the source; quotes that can't be found → "unclear"
Bifrost

→

6 · Score + expiry
Freshness, best-before, fix diff, stored for Watch
SQLite

Freshness

Q = 0.6·Recency + 0.4·Authority

supported 60+40Q · unclear 30+20Q · contradicted 15(1−Q)

Best-before

min(category half-life, measured release cadence)

unclear ÷ 3 · contradicted = expired now

SPONSOR TECHNOLOGY

Tavily finds the truth. Bifrost keeps it cheap and reliable.

Tavily

- Advanced search with **include_domains** set to the vendor's official docs, plus an open-web fallback
- **chunks_per_source** gives the judge clean, quotable evidence
- Publish dates feed the recency score
- Runs keyless out of the box; add an API key for higher limits

Bifrost by Maxim AI

- **Per-step routing**: cheap extractor and a stronger judge, any provider
- **Automatic fallback**: live "provider outage" toggle, rerouted calls shown in the UI
- **Cache** on Redis: repeated checks cost 0 tokens
- **Virtual key + budget + rate limit**; when hit, LumaSuri stops instead of guessing

Routing, cache and fallback data come from Bifrost's `extra_fields`

RESULTS

Measured, not promised.

87.5%

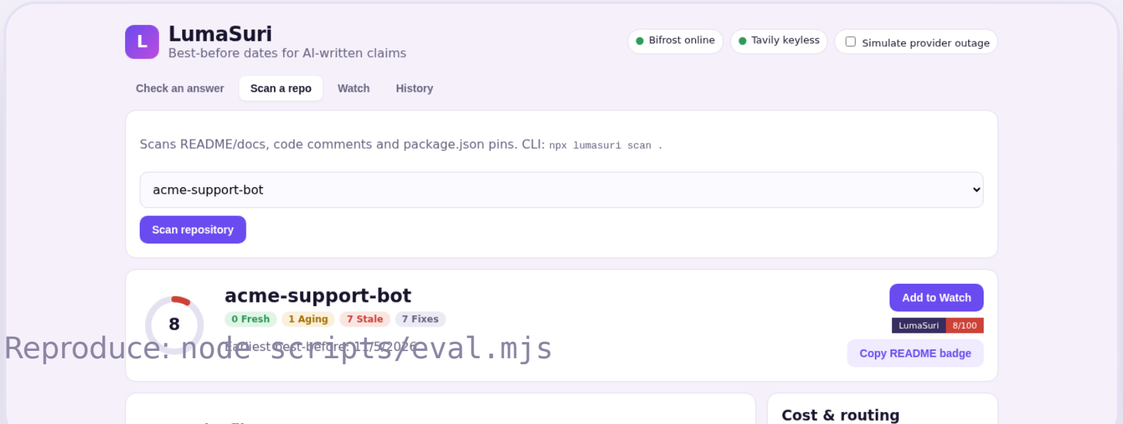
accuracy on 16 labeled current/stale claims, with a **3B local model**

10/16

claims settled by registries with **zero LLM tokens**

7

stale claims found in the demo repo (README, code comments, package.json) with only **3 LLM calls** and 7 fix diffs



Honest misses

The 2 misses ("Bard", "Twitter CEO") came from the small local judge trusting older pages. Switching the judge to a stronger model through Bifrost is one environment variable.

Reproduce: node scripts/eval.mjs

WHY IT STANDS OUT

Not just "is this true?" but "until when?"

	FACT-CHECK CHAT TOOLS	DOC-DRIFT LINTERS	LUMASURI
Checks AI answers	✓	—	✓
Checks repos (docs, comments, pins)	—	✓ (code vs docs)	✓ **(vs the live world)**
Predicts expiry (best-before)	—	—	✓ **from release cadence**
Re-checks over time + "what changed"	—	on commit	✓ **Watch**
Rejects verdicts it can't quote	rarely	n/a	✓ **evidence lock**
Zero-token deterministic path	—	partial	✓ **registries + ledger**

IMPACT & SCALE

Verify once, reuse everywhere, until it expires.

Who needs it

- Dev teams shipping AI-written docs and code
- Support bots and RAG apps (guard proxy)
- Educators and students using AI answers
- Open-source maintainers keeping READMEs honest

Why it scales

- **Claim ledger**: "Node 18 is LTS" in 1,000 repos is verified once
- Registries handle most version claims for free
- Bifrost: swap providers, add budgets, no code changes
- Runs fully local (Ollama) or in the cloud

Next

GitHub Action that posts fix diffs on PRs · browser extension for ChatGPT/Gemini · team Watch digests · more registries (Maven, crates, Go, Docker Hub)

L

LumaSuri

Know what your AI got right, and how long it stays right.

github.com/dioxaaa/lumasuri

Built for the Global AI Hackathon (HackCulture × Geek Room) · Problem Statement 06 · Tavily + Bifrost by Maxim AI